
```

fprintf('=== prob 2: sin/cos(x^2) pe [-1,1] ===\n'); prob2();
fprintf('=== prob 3: chebyshev1, w=1/sqrt(1-x^2) ===\n'); prob3();
fprintf('=== prob 4: chebyshev2, w=sqrt(1-x^2) ===\n'); prob4();
fprintf('=== prob 5: gauss-laguerre ===\n'); prob5();
fprintf('=== prob 6: gauss-hermite ===\n'); prob6();
fprintf('=== prob 7: gauss-jacobi ===\n'); prob7();

%pb1

function [x, A] = gjmat(al, be, n)
% construieste matricea jacobi tridiagonala si calculeaza noduri+ponderi
% al = alpha_0..alpha_{n-1}
% be = beta_0..beta_{n-1}
    d = al(1:n);
    e = sqrt(be(2:n));
    J = diag(d) + diag(e, 1) + diag(e, -1);
    [V, D] = eig(J);
    x = diag(D)';
    [x, ix] = sort(x);
    V = V(:, ix);
    A = be(1) * V(1,:).^2;
end

function [x, A] = gaujac(n, alp, bet)
% gauss-jacobi pe [-1,1]: w(t) = (1-t)^alp * (1+t)^bet
    al = zeros(1, n);
    be = zeros(1, n);
    ab = alp + bet;
    be(1) = 2^(ab+1) * gamma(alp+1)*gamma(bet+1) / gamma(ab+2);
    for k = 0:n-1
        num = bet^2 - alp^2;
        den = (2*k + ab) * (2*k + ab + 2);
        if abs(den) < 1e-14
            al(k+1) = 0;
        else
            al(k+1) = num / den;
        end
    end
    if n >= 2
        be(2) = 4*(1+alp)*(1+bet) / ((2+ab)^2*(3+ab));
    end
    for k = 2:n-1
        be(k+1) = 4*k*(k+alp)*(k+bet)*(k+ab) / ...
            ((2*k+ab-1)*(2*k+ab)^2*(2*k+ab+1));
    end
    [x, A] = gjmat(al, be, n);
end

function [x, A] = gauleg(n)
% gauss-legendre pe [-1,1], w=1
    [x, A] = gaujac(n, 0, 0);
end

```

```

function [x, A] = gaugeb1(n)
% gauss-chebyshev tip 1 pe [-1,1]: w=(1-t^2)^(-1/2)
[x, A] = gaujac(n, -0.5, -0.5);
end

function [x, A] = gaugeb2(n)
% gauss-chebyshev tip 2 pe [-1,1]: w=(1-t^2)^(1/2)
[x, A] = gaujac(n, 0.5, 0.5);
end

function [x, A] = gaulag(n, mu)
% gauss-laguerre pe [0,inf): w=t^mu * exp(-t), mu > -1
al = zeros(1, n);
be = zeros(1, n);
be(1) = gamma(1 + mu);
for k = 0:n-1
    al(k+1) = 2*k + mu + 1;
end
for k = 1:n-1
    be(k+1) = k * (k + mu);
end
[x, A] = gjmat(al, be, n);
end

function [x, A] = gauherm(n)
% gauss-hermite pe R: w=exp(-t^2)
al = zeros(1, n);
be = zeros(1, n);
be(1) = sqrt(pi);
for k = 1:n-1
    be(k+1) = k / 2;
end
[x, A] = gjmat(al, be, n);
end

%pb2

function prob2()
ep = 1e-7;
refs = integral(@(x) sin(x.^2), -1, 1);
refc = integral(@(x) cos(x.^2), -1, 1);
ns = 0; nc = 0;
for n = 1:100
    [xn, An] = gauleg(n);
    qs = sum(An .* sin(xn.^2));
    qc = sum(An .* cos(xn.^2));
    if ns == 0 && abs(qs - refs) < ep
        ns = n;
        fprintf(' sin(x^2): %d noduri, val=%.10f, err=%.2e\n', ...
            n, qs, abs(qs-refs));
    end
    if nc == 0 && abs(qc - refc) < ep
        nc = n;
    end
end

```

```

        fprintf('  cos(x^2): %d noduri, val=%.10f, err=%.2e\n', ...
            n, qc, abs(qc-refc));
    end
    if ns > 0 && nc > 0
        break;
    end
end
end

%pb3

function prob3()
    n = 10;
    [xn, An] = gauceb1(n);
    f = @(t) t .* exp(-t.^2);
    q = sum(An .* f(xn));
    ex = integral(@(t) t.*exp(-t.^2)./sqrt(1-t.^2), -1, 1);
    fprintf('  q   = %.10f\n', q);
    fprintf('  ref = %.10f (integranda impară => 0)\n', ex);
    fprintf('  err = %.2e\n', abs(q - ex));
end

%pb4

function prob4()
    n = 10;
    [xn, An] = gauceb2(n);
    f = @(t) exp(-t.^2);
    q = sum(An .* f(xn));
    ex = integral(@(t) sqrt(1-t.^2).*exp(-t.^2), -1, 1);
    err = abs(q - ex);
    fprintf('  q   = %.10f\n', q);
    fprintf('  ref = %.10f\n', ex);
    fprintf('  err = %.2e\n', err);
end

%pb5

function prob5()
    n = 20;
    [xn, An] = gaulag(n, 0);
    qs = sum(An .* sin(xn));
    qc = sum(An .* cos(xn));
    fprintf('  int e^-x sin(x): %.10f (exact 0.5, err=%.2e)\n', qs,
abs(qs-0.5));
    fprintf('  int e^-x cos(x): %.10f (exact 0.5, err=%.2e)\n', qc,
abs(qc-0.5));
end

%pb6

function prob6()
    n = 20;
    [xn, An] = gauherm(n);

```

```

    qs = sum(An .* sin(xn));
    qc = sum(An .* cos(xn));
    exc = sqrt(pi) * exp(-0.25);
    fprintf('  int e^-x^2 sin(x): %.10f (exact 0, err=%.2e)\n', qs, abs(qs));
    fprintf('  int e^-x^2 cos(x): %.10f (exact %.10f, err=%.2e)\n', ...
        qc, exc, abs(qc - exc));
end

%pb7

function prob7()
    % valoare exacta: gamma(1/4)*sqrt(pi) / (2*gamma(3/4))
    ex = gamma(0.25)*sqrt(pi) / (2*gamma(0.75));
    for n = [5 10 15 20]
        [tn, An] = gaujac(n, 0, -0.5);

        arg = pi*(1 + tn)/4;
        h = (pi/4) * sqrt((1 + tn) ./ sin(arg));
        q = sum(An .* h);
        err = abs(q - ex);
        fprintf('  n=%2d: val=%.10f, exact=%.10f, err=%.2e\n', n, q, ex,
err);
    end
end

=== prob 2: sin/cos(x^2) pe [-1,1] ===
    sin(x^2): 7 noduri, val=0.6205366361, err=3.26e-08
    cos(x^2): 7 noduri, val=1.8090484594, err=1.64e-08
=== prob 3: chebyshev1, w=1/sqrt(1-x^2) ===
    q = -0.0000000000
    ref = -0.0000000000 (integranda impara => 0)
    err = 9.71e-16
=== prob 4: chebyshev2, w=sqrt(1-x^2) ===
    q = 1.2589242566
    ref = 1.2589242566
    err = 2.63e-13
=== prob 5: gauss-laguerre ===
    int e^-x sin(x): 0.5000000000 (exact 0.5, err=1.79e-14)
    int e^-x cos(x): 0.5000000000 (exact 0.5, err=7.73e-14)
=== prob 6: gauss-hermite ===
    int e^-x^2 sin(x): 0.0000000000 (exact 0, err=2.97e-16)
    int e^-x^2 cos(x): 1.3803884470 (exact 1.3803884470, err=2.22e-16)
=== prob 7: gauss-jacobi ===
    n= 5: val=2.6220575391, exact=2.6220575543, err=1.52e-08
    n=10: val=2.6220575543, exact=2.6220575543, err=1.78e-15
    n=15: val=2.6220575543, exact=2.6220575543, err=4.44e-16
    n=20: val=2.6220575543, exact=2.6220575543, err=4.44e-16

```